Middle East Technical University
Department of Electrical and Electronics Engineering

# EE201 – Circuit Theory I

## Lecture 6: Dependent Sources and the Ideal Transformer

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| Colour code used in these notes and in the slides | | | |
|---|---|---|---|
| current $i$ | voltage $v$ | power $p$ | magnetic flux $\Phi$ |

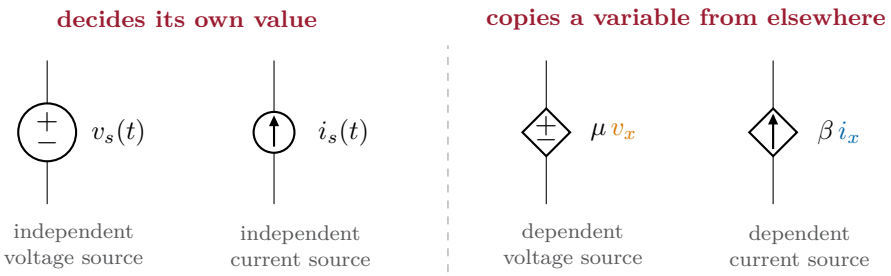
## 1. Sources controlled from elsewhere

The independent sources met so far decide their own value. A 12 V source holds 12 V whatever the rest of the circuit does. The sources of this lecture do not: their value is set by a voltage or a current *somewhere else* in the circuit. They are how amplifiers, transistors and many integrated circuits enter a circuit diagram.

**Dependent (controlled) source**

A **dependent source** is a voltage or current source whose value is a function of a voltage or current in *another* branch of the circuit, the **controlling variable**. Almost always the function is a constant times one controlling variable, as in $v = \mu\, v_x$: a **linear** dependent source. More general forms are in Section 2.2.

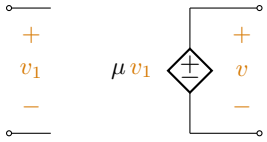
The symbol is a **diamond**. A circle means the source decides its own value; a diamond means it copies, scaled, a variable from elsewhere.




## 2. The four kinds

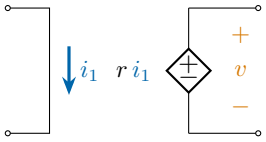
A dependent source senses a voltage or a current, and produces a voltage or a current. That gives four kinds. The name reads in that order — what it senses, then what it produces.

**1. VCVS — voltage-controlled voltage source**



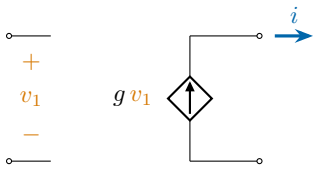

$v = \mu\, v_1$

**2. CCVS — current-controlled voltage source**



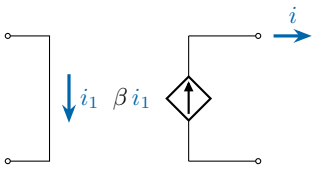

$v = r\, i_1$

**3. VCCS — voltage-controlled current source**




$i = g\, v_1$

**4. CCCS — current-controlled current source**




$i = \beta\, i_1$

| Kind | Terminal equation | The constant is called | Unit |
|---|---|---|---|
| VCVS | $v = \mu\, v_1$ | voltage gain | none (V/V) |
| CCVS | $v = r\, i_1$ | transresistance | ohm, Ω (V/A) |
| VCCS | $i = g\, v_1$ | transconductance | siemens, S (A/V) |
| CCCS | $i = \beta\, i_1$ | current gain | none (A/A) |

The unit of the constant is always "what comes out" over "what goes in". When the two are different kinds of quantity the name carries *trans-*: a transresistance turns a current into a voltage, so it is measured in ohms, like a resistance.

### 2.1. The sensing side

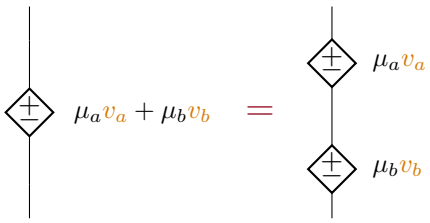
In each picture the left-hand pair of terminals is where the source senses its controlling variable. Sensing does not change the circuit — it does not "disturb" it:

- to sense a voltage it is an **open circuit** — it draws no current;
- to sense a current it is a **short circuit** — it takes no voltage.

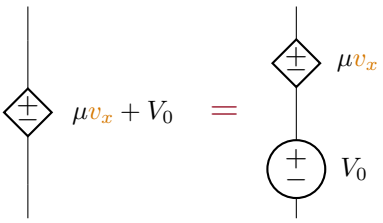
In circuit diagrams the sensing side is usually not drawn: the controlling variable is labelled on an element that is already there, as $v_x$, $i_x$ or $i_1$.

### 2.2. More general dependent sources

The four kinds above depend on one variable through one constant. That covers almost every circuit in this course, but the definition allows more.




several variables: two sources in series

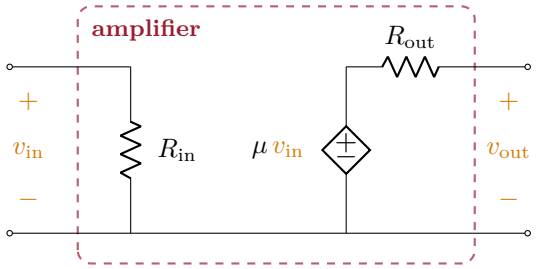



affine: the constant is an independent source

- **Several controlling variables**, $v = \mu_a v_a + \mu_b v_b$: the same as two dependent sources in series, one giving $\mu_a v_a$ and one giving $\mu_b v_b$. For a current source the two are in parallel.
- **Affine**, $v = \mu v_x + V_0$: the constant term is an independent source in series with a linear dependent one. Such sources may appear in problems.
- **Nonlinear**, for example $i = f(v_x)$: a real transistor over its full range behaves like this. We do not use nonlinear dependent sources in EE201.

## 3. Why we need them

A dependent source is the simplest model of anything that *amplifies*. The picture below is how an amplifier usually appears in a circuit diagram.




$R_{\text{in}}$ is the **input resistance**: the amplifier draws a small current through it. $R_{\text{out}}$ is the **output resistance**: the output acts like a source $\mu\, v_{\text{in}}$ behind it. The gain $\mu$ is the only place where the output can grow larger than the input.

Many components and integrated circuits are modelled as dependent sources over their working range, and some are designed to be one. Each of the four kinds has a familiar example:

| Kind | Device | Typical constant |
|---|---|---|
| VCVS | op-amp | $\mu \approx 10^5$ |
| CCVS | current-sense amplifier | $r$, in ohms |
| VCCS | MOSFET; transconductance amplifier | $g_m$, in siemens |
| CCCS | bipolar transistor | $\beta \approx 100$ |

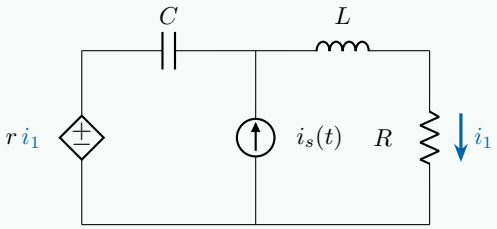
The op-amp has a chapter of its own later in this course; the transistor models come in the electronics courses.

**DC/DC converters.** A DC/DC converter turns one DC voltage into another — the 5 V of a USB port into the 1 V a processor needs, a battery voltage into what a motor needs. It is one of the most fundamental electronic circuits, used in almost every field: phones, computers, cars, satellites, robots. To the circuit around it, a DC/DC converter is essentially a voltage-controlled voltage source, $v_{\text{out}} = k\, v_{\text{in}}$, with the ratio $k$ set by its design.

## 4. Reading a diagram with a dependent source

**Example 6.1 — the controlling variable of a CCVS**

Which equation does the dependent source contribute in this circuit?




It is a CCVS, so its terminal equation is $v = r\, i_1$ — and $i_1$ is the current through $R$. The source sits at the far left; the current it senses flows at the far right.

Count as before: $n = 4$ nodes and $b = 5$ branches, so $2b = 10$ unknowns and $b = 5$ terminal equations. Four of them — for $C$, $L$, $i_s$ and $R$ — each speak about one element. The fifth ties the source's own voltage to another branch's current.

**What is new**

So far every terminal equation described one element *in isolation*. A dependent source does not: its terminal equation names a variable of another branch. Nothing else changes. KCL and KVL are the same, and there are still $2b$ equations for $2b$ unknowns.

**Two meanings of "controlled"**

Earlier, a resistor was called *voltage-controlled* if its current is a function of its *own* voltage, $i = f(v)$. A voltage-controlled *source* is different: its value is set by the voltage of *another* branch. The first describes one element's curve; the second is a link between two branches.

## 5. Solving circuits with dependent sources

No new method is needed.

**Recipe**

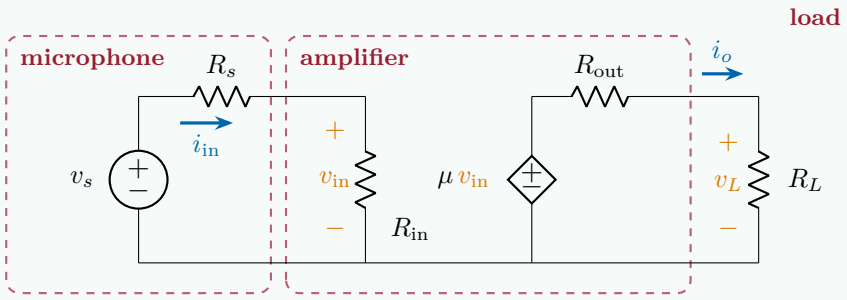
1. Treat the dependent source as if it were independent, with its value written as $\mu v_x$, $r i_x$, $g v_x$ or $\beta i_x$.
2. Write KCL, KVL and the terminal equations as usual.
3. Add one equation that gives the controlling variable in terms of your unknowns.

**Example 6.2 — an amplifier**

A microphone drives the amplifier of Section 3, which drives a load.

- **Microphone:** $v_s = 50\,\text{mV}$, with internal resistance $R_s = 1\,\text{k}\Omega$.
- **Amplifier:** input resistance $R_{\text{in}} = 4\,\text{k}\Omega$, gain $\mu = 100$, output resistance $R_{\text{out}} = 100\,\Omega$.
- **Load:** $R_L = 400\,\Omega$.

Find $v_L$, and compare the power the microphone supplies with the power the load receives.




**Input side.** One loop, a divider: $i_{\text{in}} = 50\,\text{mV}/5\,\text{k}\Omega = 10\,\mu\text{A}$, so $v_{\text{in}} = 4\,\text{k}\Omega \times 10\,\mu\text{A} = 40\,\text{mV}$.
**Output side.** The dependent source now has a known value: $100 \times 40\,\text{mV} = 4\,\text{V}$. Another divider: $i_o = 4\,\text{V}/500\,\Omega = 8\,\text{mA}$, so

$$\boxed{v_L = 400\,\Omega \times 8\,\text{mA} = 3.2\,\text{V}}, \qquad \frac{v_L}{v_s} = 64.$$

The overall gain is 64, not 100: part of the signal is lost in $R_s$ and part in $R_{\text{out}}$.

## Example 6.2, continued — where the power comes from

With the PSC on every element:

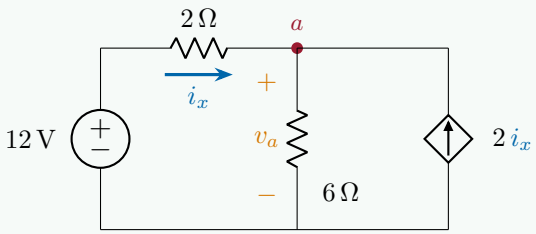
| input side | | output side | |
|---|---|---|---|
| microphone $v_s$ | $-0.5\,\mu$W | dependent source | $-32$ mW |
| $R_s$ | $+0.1\,\mu$W | $R_{\text{out}}$ | $+6.4$ mW |
| $R_{\text{in}}$ | $+0.4\,\mu$W | $R_L$ | $+25.6$ mW |

Each side sums to zero. The load receives 25.6 mW while the microphone supplies only 0.5 $\mu$W — 51 200 times less. The dependent source supplied the difference. A real amplifier takes that energy from its power supply, which the model does not draw.

One more detail: the bottom wire is shared, yet it carries no current. Wrap a closed surface around the input side: that wire is the only branch crossing it, so by KCL its current is zero. The two sides interact only through $v_{\text{in}}$.

## Example 6.3 — when the controlling variable is unknown

The CCCS pushes $2i_x$ into node $a$, where $i_x$ is the current through the 2 Ω resistor. Find $i_x$ and $v_a$.




**Equations.** Here step 3 is free: $i_x$ is already one of the unknowns.

$$\text{KVL:}\quad -12+2i_x+v_a=0, \qquad \text{KCL at } a\text{:}\quad -i_x-2i_x+\frac{v_a}{6}=0 \implies v_a=18\,i_x.$$

**Solution.** $-12+2i_x+18i_x=0$, so

$$\boxed{i_x=0.6\ \text{A}}, \qquad \boxed{v_a=10.8\ \text{V}}.$$

**Power check.** The 12 V source supplies 7.2 W. The resistors absorb $2(0.6)^2=0.72$ W and $10.8^2/6=19.44$ W. The CCCS carries 1.2 A *out of* its + terminal, so it absorbs $10.8\times(-1.2)=-12.96$ W:

$$-7.2+0.72+19.44-12.96=0.$$

The 6 Ω resistor absorbs more than the independent source supplies. The dependent source makes up the rest.

**What a dependent source cannot do**

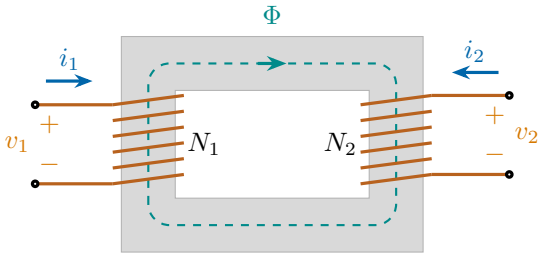
**It cannot start anything.** Switch the 12 V source of Example 6.3 off. The equations become $2i_x + v_a = 0$ and $v_a = 18i_x$, so $20\,i_x = 0$: every voltage and current is zero, the dependent source's included. Without an independent source, all-zero satisfies every equation — and when the circuit has a unique solution, that is the solution. (The constant term of an affine source, $\mu v_x + V_0$, is an independent source in disguise; that part can.)

**It is never switched off.** Later we will switch sources off one at a time (superposition, Thévenin equivalents). Only *independent* sources are switched off. A dependent source stays: its value is not an input, it is part of how the circuit behaves.

## 6. The ideal transformer

So far we have assumed that no changing magnetic field links the loops of a circuit. Sometimes we need such a link on purpose. We then model it as an element of its own: the transformer.

A transformer is two coils wound on one iron core. A current in coil 1 sets up a magnetic field, and the core guides that field through coil 2. When the current changes, the field changes, and the changing field induces a voltage in coil 2. No wire connects the two coils: the magnetic field links them.




The iron core guides the magnetic field of coil 1 (flux $\Phi$) through coil 2.

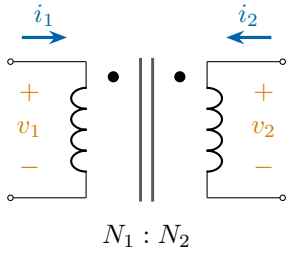
A voltage is induced only while the field changes, so a physical transformer works only with **changing (AC) signals**. A constant current gives a constant field, and no voltage appears on the other coil.

**Ideal transformer**

With $v_1$, $v_2$ positive at the dotted terminals and $i_1$, $i_2$ entering the dotted terminals,

$$\boxed{\frac{v_1}{N_1} = \frac{v_2}{N_2}} \qquad \boxed{N_1 i_1 + N_2 i_2 = 0}$$

Equivalently, $v_2 = \dfrac{N_2}{N_1}\, v_1$ and $i_2 = -\dfrac{N_1}{N_2}\, i_1$.




The two lines are the core. The **dots** act like reference marks: move one dot to the other end of its winding and both equations change sign. The equations have no $d/dt$, so in the model the ideal transformer is a *resistive* element, like resistors and dependent sources.

## 7. The power of an ideal transformer

Each winding has its current entering its + terminal, so $v_1 i_1$ and $v_2 i_2$ are the powers the two windings absorb.

**The ideal transformer absorbs no power**

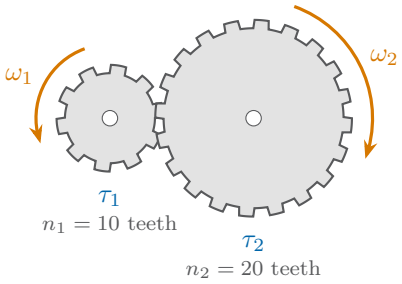
At every instant, $p(t) = v_1 i_1 + v_2 i_2 = 0$.

**Proof.** Substitute the two equations: $v_2 i_2 = (\frac{N_2}{N_1} v_1)(-\frac{N_1}{N_2} i_1) = -v_1 i_1$. ■

Nothing is stored and nothing is dissipated: the voltage is multiplied by $N_2/N_1$ and the current divided by the same factor. The same holds for any number of windings on one core.

### 7.1. A mechanical analogy: the gear pair

An ideal gear pair does for rotation what an ideal transformer does for electric circuits. Take angular velocity $\omega$ as the across quantity, like voltage, and torque $\tau$ as the through quantity, like current.

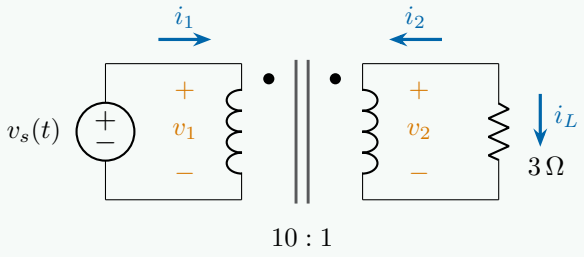



| Ideal transformer | Ideal gear pair |
|---|---|
| voltage $v$ | angular velocity $\omega$ |
| current $i$ | torque $\tau$ |
| turns $N_1$, $N_2$ | teeth $n_1$, $n_2$ |
| power in = power out | $\tau_1\omega_1 = \tau_2\omega_2$ |
| step-down: $v$ down, $i$ up | small drives large: $\omega$ down, $\tau$ up |
| moving a dot flips a sign | meshing gears turn opposite ways |

With $n_1 = 10$ and $n_2 = 20$ teeth, the large gear turns at half the speed with twice the torque — what a $2:1$ step-down transformer does to voltage and current. Both are ideal only in the model: no losses and nothing stored (the gears are massless and frictionless). Unlike a physical transformer, a gear pair also works at constant speed.

**Example 6.4 — a step-down transformer**

A $10:1$ transformer connects $v_s(t) = 120\cos\omega t$ V to a $3\,\Omega$ load. Find the currents and the power on each side.




$v_2 = \frac{1}{10} v_1 = 12\cos\omega t$ V. The load takes $i_L = v_2/3 = 4\cos\omega t$ A, which leaves the dotted terminal, so $i_2 = -4\cos\omega t$ A and $i_1 = -\frac{1}{10} i_2 = 0.4\cos\omega t$ A.

$$p_1 = v_1 i_1 = 48\cos^2\omega t\,\text{W}, \qquad p_2 = v_2 i_2 = -48\cos^2\omega t\,\text{W}, \qquad p_1 + p_2 = 0.$$

Seen from the source, $v_1/i_1 = 300\,\Omega = 10^2 \times 3\,\Omega$: a transformer rescales a resistance by $(N_1/N_2)^2$.

## Summary

- A **dependent source** (diamond) takes its value from a voltage or current in another branch. Four kinds: VCVS, CCVS, VCCS, CCCS; the constant's unit is output over input. Almost always linear in one variable; several variables and affine forms split into simpler sources.
- Its terminal equation names a **foreign variable**. Otherwise nothing changes: write KCL, KVL and terminal equations, plus one equation for the controlling variable.

- It can **deliver power** — that is what amplification is — but it **cannot start anything**: without an independent source everything is zero. It is never switched off.
- The **ideal transformer**: $v_1/N_1 = v_2/N_2$ and $N_1 i_1 + N_2 i_2 = 0$, signs set by the **dots**. It absorbs **no power at any instant**. A physical one works only with AC signals.

### Next lecture

From circuit to **graph**: labelling the nodes and branches of a circuit so that its $2b$ equations can be written systematically — the first step towards node and mesh analysis.

### Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Section 1.6 (dependent sources) and Section 13.5 (ideal transformers).
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits* — controlled sources and the ideal transformer.
- Lecture notes and videos of Prof. Emre Tuna, `users.metu.edu.tr/etuna/ee201` — notes p. 11; Lecture 5 video: dependent sources 41:47, the four kinds 43:04–48:14, example 49:15, ideal transformer 51:31, its power 54:47.